

# Cumulative Incidence Curves

## 1.1 Setup

Consider time to event data  $\{(T_i, \delta_i)\}$ , where time  $T_i$  is absolutely continuous and the status  $\delta_i$  is coded as:

$$\delta_i = \begin{cases} 0 & \text{censoring,} \\ 1 & \text{event,} \\ 2 & \text{competing risk.} \end{cases}$$

**Definition 1.1.** The **cumulative incidence curve** (CIC) for type-1 events  $F_1(t)$  is the probability of experiencing the event before the competing risk by time  $t$ :

$$F_1(t) = \mathbb{P}(T \leq t, \delta = 1) = \int_0^t S(u) \lambda_1(u) du,$$

where  $S$  is the overall survival function:

$$S(t) = \mathbb{P}(T > t) = e^{-\Lambda_1(t) - \Lambda_2(t)}.$$

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**Discussion 1.1.** The overall survival function  $S(t)$  may be estimated from data of the form  $\{(T_i, \delta_i^*)\}$ , where:

$$\delta_i^* = \begin{cases} 0 & \text{censoring,} \\ 1 & \text{event or competing risk.} \end{cases}$$

Let  $\hat{S}$  denote the Kaplan-Meier estimate of  $S$ . The Nelson-Aalen estimate of  $\Lambda_1(u)$  is:

$$\hat{\Lambda}_1(t) = \int_0^t \frac{dN_1(u)}{Y(u)},$$

where  $Y(u) = \sum_{i=1}^n \mathbb{I}(T_i \geq u)$  is the number at risk and  $N_1(u)$  is the counting process for type-1 events:

$$N_1(t) = \sum_{i=1}^n \mathbb{I}(T_i \leq t, \delta_i = 1).$$

The standard estimator of  $F_1$  is:

$$\hat{F}_1(t) = \int_0^t \hat{S}(u-) d\hat{\Lambda}_1(u),$$

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## 1.2 Background

We make use of the following results proved elsewhere.

**Proposition 1.1 (Kaplan-Meier to Nelson-Aalen).** Let  $S(t)$  denote the overall survival function and  $\Lambda(t) = \Lambda_1(t) + \Lambda_2(t)$  the overall cumulative hazard. Then:

$$\frac{\sqrt{n}\{\hat{S}(t) - S(t)\}}{-S(t)} = \sqrt{n}\{\hat{\Lambda}(t) - \Lambda(t)\} + o_p(1), \quad (1.1)$$

where  $\hat{S}$  is the KM estimator, and  $\hat{\Lambda}$  the NA. ◆

**Proposition 1.2 (Martingale Expansion of Nelson-Aalen).** The NA estimator is expressible as:

$$\sqrt{n}\{\hat{\Lambda}(t) - \Lambda(t)\} = \int_0^t \frac{\sqrt{n}}{Y(u)} dM(u) + o_p(1),$$

where  $M(u) = N(u) - \int_0^u Y(s) d\Lambda(s)$  is the counting process martingale.

Likewise, for the cause-specific cumulative hazard:

$$\sqrt{n}\{\hat{\Lambda}_1(t) - \Lambda_1(t)\} = \int_0^t \frac{\sqrt{n}}{Y(u)} dM_1(u) + o_p(1),$$

where  $M_1(u) = N_1(u) - \int_0^u Y(s) d\Lambda_1(s)$  is the counting process martingale. ◆

**Proposition 1.3.** The predictable covariation is a bilinear form:

$$\langle M_1 + M_2, M_1 + M_2 \rangle = \langle M_1 \rangle + \langle M_2 \rangle + 2\langle M_1, M_2 \rangle.$$

The analogous result holds for the optional covariation. ◆

**Proposition 1.4.** If  $H_1, H_2$  are predictable processes and  $M_1, M_2$  are mean-zero martingales, then the predictable covariation of the corresponding stochastic integrals is:

$$\left\langle \int H_1(t) dM_1(t), \int H_2(t) dM_2(t) \right\rangle = \int H_1(t) H_2(t) d\langle M_1(t), M_2(t) \rangle.$$

The analogous result holds for the optional covariation. ◆

## 1.3 Martingale Representation

**Proposition 1.5.**

$$\begin{aligned} \sqrt{n}\{\hat{F}_1(t) - F_1(t)\} &= \int_0^t \sqrt{n}\{\hat{S}(u) - S(u)\} d\hat{\Lambda}_1(u) \\ &\quad + \int_0^t \sqrt{n}S(u)\{d\hat{\Lambda}_1(u) - d\Lambda_1(u)\}. \end{aligned}$$

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**Proof.**

$$\sqrt{n}\{\hat{F}_1(t) - F_1(t)\} = \int_0^t \sqrt{n}\hat{S}(u)d\hat{\Lambda}_1(u) - \int_0^t \sqrt{n}S(u)d\Lambda_1(u).$$

Adding and subtracting  $S(u)$  from the first integrand:

$$\begin{aligned} \int_0^t \sqrt{n}\hat{S}(u)d\hat{\Lambda}_1(u) &= \int_0^t \sqrt{n}\{\hat{S}(u) - S(u) + S(u)\}d\hat{\Lambda}_1(u) \\ &= \int_0^t \sqrt{n}\{\hat{S}(u) - S(u)\}d\hat{\Lambda}_1(u) + \int_0^t \sqrt{n}S(u)d\hat{\Lambda}_1(u). \end{aligned}$$

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**Proposition 1.6.**

$$\begin{aligned} \int_0^t \sqrt{n}\{\hat{S}(u) - S(u)\}d\hat{\Lambda}_1(u) &= - \int_0^t \sqrt{n}\{\hat{\Lambda}(u) - \Lambda(u)\} \cdot dF_1(u) + o_p(1) \\ &= -\sqrt{n}\{\hat{\Lambda}(t) - \Lambda(t)\}F_1(t) + \int_0^t \sqrt{n}F_1(u)d\{\hat{\Lambda}(u) - \Lambda(u)\} + o_p(1). \end{aligned}$$

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**Proof.** Since  $\hat{\Lambda}_1(\cdot)$  is consistent for  $\Lambda_1(\cdot)$ :

$$\int_0^t \sqrt{n}\{\hat{S}(u) - S(u)\}d\hat{\Lambda}_1(u) = \int_0^t \sqrt{n}\{\hat{S}(u) - S(u)\}d\Lambda_1(u) + o_p(1).$$

Applying (1.1):

$$\begin{aligned} \int_0^t \sqrt{n}\{\hat{S}(u) - S(u)\}d\Lambda_1(u) &= - \int_0^t \frac{\sqrt{n}\{\hat{S}(u) - S(u)\}}{-S(u)} \cdot S(u)d\Lambda_1(u) \\ &= - \int_0^t \sqrt{n}\{\hat{\Lambda}(u) - \Lambda(u)\} \cdot dF_1(u) + o_p(1). \end{aligned}$$

Integrating by parts:

$$\begin{aligned} \int_0^t \sqrt{n}\{\hat{\Lambda}(u) - \Lambda(u)\} \cdot dF_1(u) &= \sqrt{n} \left[ \{\hat{\Lambda}(u) - \Lambda(u)\}F_1(u) \right]_{u=0}^{u=t} - \int_0^t \sqrt{n}F_1(u)d\{\hat{\Lambda}(u) - \Lambda(u)\} \\ &= \sqrt{n}\{\hat{\Lambda}(t) - \Lambda(t)\}F_1(t) - \int_0^t \sqrt{n}F_1(u)d\{\hat{\Lambda}(u) - \Lambda(u)\}. \end{aligned}$$

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**Proposition 1.7 (Martingale Representation).**

$$\begin{aligned}\sqrt{n}\{\hat{F}_1(t) - F_1(t)\} &= -F_1(t) \int_0^t \frac{\sqrt{n} \cdot dM(u)}{Y(u)} + \int_0^t \frac{\sqrt{n} \cdot F_1(u)}{Y(u)} dM(u) \\ &\quad + \int_0^t \frac{\sqrt{n} \cdot S(u)}{Y(u)} dM_1(u) + o_p(1).\end{aligned}$$

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**Proof.** From the previous propositions:

$$\begin{aligned}\sqrt{n}\{\hat{F}_1(t) - F_1(t)\} &= -\sqrt{n}\{\hat{\Lambda}(t) - \Lambda(t)\}F_1(t) + \int_0^t \sqrt{n}F_1(u)d\{\hat{\Lambda}(u) - \Lambda(u)\} \\ &\quad + \int_0^t \sqrt{n}S(u)\{d\hat{\Lambda}_1(u) - d\Lambda_1(u)\} + o_p(1).\end{aligned}$$

The conclusion follows from applying the martingale expansion of the Nelson-Aalen estimator to  $\sqrt{n}\{\hat{\Lambda}(t) - \Lambda(t)\}$ ,  $\sqrt{n}d\{\hat{\Lambda}(t) - \Lambda(t)\}$ , and  $\sqrt{n}d\{\hat{\Lambda}_1(t) - \Lambda_1(t)\}$ .

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**Corollary 1.1.**

$$\begin{aligned}\sqrt{n}\{\hat{F}_1(t) - F_1(t)\} &= -F_1(t) \int_0^t \frac{\sqrt{n}}{Y(u)} dM_1(u) + \int_0^t \frac{\sqrt{n} \cdot \{1 - F_2(u)\}}{Y(u)} dM_1(u) \\ &\quad - F_1(t) \int_0^t \frac{\sqrt{n}}{Y(u)} dM_2(u) + \int_0^t \frac{\sqrt{n} \cdot F_1(u)}{Y(u)} dM_2(u) + o_p(1).\end{aligned}$$

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**Proof.** Substituting  $M(u) = M_1(u) + M_2(u)$  and  $S(u) = 1 - F_1(u) - F_2(u)$  gives the result.

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**1.4 Predictable and Optional Variations****Proposition 1.8.** Suppose  $n^{-1}Y(t) \xrightarrow{p} y(t)$ , then as  $n \rightarrow \infty$ :

$$\begin{aligned}\langle \sqrt{n}\{\hat{F}_1(t) - F_1(t)\} \rangle &\xrightarrow{p} F_1^2(t) \int_0^t \frac{\lambda_1(u)}{y(u)} du + \int_0^t \frac{\{1 - F_2(u)\}^2 \lambda_1(u)}{y(u)} du \\ &\quad + F_1^2(t) \int_0^t \frac{\lambda_2(u)}{y(u)} du + \int_0^t \frac{F_1^2(u) \lambda_2(u)}{y(u)} du \\ &\quad - 2F_1(t) \int_0^t \frac{\{1 - F_1(u)\} \lambda_1(u)}{y(u)} du \\ &\quad - 2F_1(u) \int_0^t \frac{F_1(u) \lambda_2(u)}{y(u)} du\end{aligned}\tag{1.2}$$

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**Proof.** Finding the optional variation:

$$\begin{aligned} \langle \sqrt{n} \{ \hat{F}_1(t) - F_1(t) \} \rangle &= F_1^2(t) \int_0^t \frac{n}{Y^2(u)} d\langle M_1(u) \rangle + \int_0^t \frac{n \{1 - F_2(u)\}^2}{Y^2(u)} d\langle M_1(u) \rangle \\ &\quad + F_1^2(t) \int_0^t \frac{n}{Y^2(u)} d\langle M_2(u) \rangle + \int_0^t \frac{n F_1^2(u)}{Y^2(u)} d\langle M_2(u) \rangle \\ &\quad - 2F_1(t) \int_0^t \frac{n \{1 - F_1(u)\}}{Y^2(u)} d\langle M_1(u) \rangle \\ &\quad - 2F_1(t) \int_0^t \frac{n F_1(u)}{Y^2(u)} d\langle M_2(u) \rangle + o_p(1). \end{aligned}$$

Substituting  $d\langle M_j(u) \rangle = Y(u) \lambda_j(u) du$ :

$$\begin{aligned} \langle \sqrt{n} \{ \hat{F}_1(t) - F_1(t) \} \rangle &= F_1^2(t) \int_0^t \frac{\lambda_1(u)}{n^{-1}Y(u)} du + \int_0^t \frac{\{1 - F_2(u)\}^2 \lambda_1(u)}{n^{-1}Y(u)} du \\ &\quad + F_1^2(t) \int_0^t \frac{\lambda_2(u)}{n^{-1}Y(u)} du + \int_0^t \frac{F_1^2(u) \lambda_2(u)}{n^{-1}Y(u)} du \\ &\quad - 2F_1(t) \int_0^t \frac{\{1 - F_2(u)\} \lambda_1(u)}{n^{-1}Y(u)} du \\ &\quad - 2F_1(t) \int_0^t \frac{F_1(u) \lambda_2(u)}{n^{-1}Y(u)} du + o_p(1). \end{aligned}$$

Letting  $n^{-1}Y(u) \xrightarrow{p} y(u)$  gives the result. ■

**Discussion 1.2.** By the martingale central limit theorem:

$$\sqrt{n} \{ \hat{F}_1(t) - F_1(t) \} \rightsquigarrow W \{ \sigma_{\text{CIC}}^2(t) \},$$

where  $\sigma_{\text{CIC}}^2(t)$  is the RHS of (1.2). An estimate of the variance is obtained by finding the optional variation:

$$\begin{aligned} \hat{\sigma}_{\text{CIC}}^2(t) &= \left[ \sqrt{n} \{ \hat{F}_1(t) - F_1(t) \} \right] = F_1^2(t) \int_0^t \frac{dN_1(u)}{n^{-1}Y^2(u)} + \int_0^t \frac{\{1 - F_2(u)\}^2 dN_1(u)}{n^{-1}Y^2(u)} \\ &\quad + F_1^2(t) \int_0^t \frac{dN_2(u)}{n^{-1}Y^2(u)} + \int_0^t \frac{F_1^2(u) dN_2(u)}{n^{-1}Y^2(u)} \\ &\quad - 2F_1(t) \int_0^t \frac{\{1 - F_2(u)\} dN_1(u)}{n^{-1}Y^2(u)} \\ &\quad - 2F_1(t) \int_0^t \frac{F_1(u) dN_2(u)}{n^{-1}Y^2(u)}. \end{aligned}$$

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## Area Under the Cumulative Incidence Curve

**Example 2.1.** Consider the area under the cumulative incidence curve (AUCIC):

$$U(\tau) = \int_0^\tau F_1(t)dt = \int_0^\tau \int_0^t S(u)d\Lambda_1(u).$$

and let  $\hat{U}(\tau)$  denote the estimator:

$$\hat{U}(\tau) = \int_0^\tau \hat{F}_1(t)dt = \int_0^\tau \int_0^t \hat{S}(u-)d\hat{\Lambda}_1(u).$$

Using the martingale representation of  $\sqrt{n}\{\hat{F}_1(t) - F_1(t)\}$ , the standardized difference:

$$\sqrt{n}\{\hat{U}(\tau) - U(\tau)\} = \int_0^\tau \sqrt{n}\{\hat{F}_1(t) - F_1(t)\}dt$$

is expressible as:

$$\begin{aligned} \int_0^\tau \sqrt{n}\{\hat{F}_1(t) - F_1(t)\}dt &= - \int_0^\tau F_1(t) \int_0^t \frac{\sqrt{n}}{Y(u)} dM_1(u)dt + \int_0^\tau \int_0^t \frac{\sqrt{n}\{1 - F_2(u)\}}{Y(u)} dM_1(u)dt \\ &\quad - \int_0^\tau F_1(t) \int_0^t \frac{\sqrt{n}}{Y(u)} dM_2(u)dt + \int_0^\tau \int_0^t \frac{\sqrt{n}F_1(u)}{Y(u)} dM_2(u)dt + o_p(1). \end{aligned}$$

The predictable variation is:

$$\begin{aligned} \langle \sqrt{n}\{\hat{U}(\tau) - U(\tau)\} \rangle &= \int_0^\tau F_1^2(t) \int_0^t \frac{\lambda_1(u)}{n^{-1}Y(u)} du dt + \int_0^\tau \int_0^t \frac{\{1 - F_2(u)\}^2 \lambda_1(u)}{n^{-1}Y(u)} du dt \\ &\quad + \int_0^\tau F_1^2(t) \int_0^t \frac{\lambda_2(u)}{n^{-1}Y(u)} du dt + \int_0^\tau \int_0^t \frac{F_1^2(u) \lambda_2(u)}{n^{-1}Y(u)} du dt \\ &\quad - 2 \int_0^\tau F_1(t) \int_0^t \frac{\{1 - F_2(u)\} \lambda_1(u)}{n^{-1}Y(u)} du dt \\ &\quad - 2 \int_0^\tau F_1(t) \int_0^t \frac{F_1(u) \lambda_2(u)}{n^{-1}Y(u)} du dt + o_p(1). \end{aligned}$$

Let  $\sigma_{\text{AUCIC}}^2(\tau)$  denote the limit in probability of the predictable variation as  $n \rightarrow \infty$ . By the martingale central limit theorem:

$$\sqrt{n}\{\hat{U}(\tau) - U(\tau)\} \rightsquigarrow W\{\sigma_{\text{AUCIC}}^2(t)\}.$$

An estimate of the asymptotic variance is obtained from the optional variation:

$$\begin{aligned} \hat{\sigma}_{\text{AUCIC}}^2(\tau) &= \int_0^\tau \hat{F}_1^2(t) \int_0^t \frac{dN_1(u)}{n^{-1}Y^2(u)} dt + \int_0^\tau \int_0^t \frac{\{1 - \hat{F}_2(u)\}^2 dN_1(u)}{n^{-1}Y^2(u)} dt \\ &\quad + \int_0^\tau \hat{F}_1^2(t) \int_0^t \frac{dN_2(u)}{n^{-1}Y^2(u)} dt + \int_0^\tau \int_0^t \frac{\hat{F}_1^2(u) dN_2(u)}{n^{-1}Y^2(u)} dt \\ &\quad - 2 \int_0^\tau \hat{F}_1(t) \int_0^t \frac{\{1 - \hat{F}_2(u)\} dN_1(u)}{n^{-1}Y^2(u)} dt \\ &\quad - 2 \int_0^\tau \hat{F}_1(t) \int_0^t \frac{\hat{F}_1(u) dN_2(u)}{n^{-1}Y^2(u)} dt. \end{aligned}$$

